

Q1. What type of relationship is defined as one resource existing only if another parent resource exist-for example, pages in a book?

- ☐ Partial
- ☒ dependent
- ☐ associative
- ☐ linked

Q2. Which URL pattern is recommended when working with one resource and a collection of resources?

- ☐ /companies/{id} and /company
- ☐ /company/{id} and /companies
- ☒ /companies/{id} and /companies
- ☐ /company/{id} and /company

Q3. When dealing with JSON web Tokens (JWTs), what is a claim?

- ☒ data in the token
- ☐ Ownership
- ☐ a permission
- ☐ and integer

Q4. Which REST constraint specifies that knowledge and understanding obtained from one component of the API should be generally applicable elsewhere in the API?

- ☒ Uniform Interface
- ☐ Client-Server
- ☐ Stateless
- ☐ Cacheable

Q5. What would you enable to allow a browser on another site to make an AJAX request to your API?

- ☐ HTTP
- ☐ REST
- ☐ OPTIONS
- ☒ CORS

Q6. APIs commonly use webhooks to _.

- ☒ notify other systems of an event
- ☐ catch error faster
- ☐ improve error logging
- ☐ log additional data

Q7. What is the underlying goal of all APIs?

- ☐ to add new technologies to an organization's infrastructure.
- ☒ to share features and functionality with other system.
- ☐ to move infrastructure to the cloud.
- ☐ to appease the latest digital transformation effort.

Q8. Which is a common command-line tool for using or exploring an API?

- ☐ bash
- ☒ curl
- ☐ ssh
- ☐ powerShell

Q9. What is the modern specification for describing an API?

- ☒ OpenAPI (Swagger)
- ☐ WADL
- ☐ WSDL
- ☐ OAuth

[OpenAPI Specification](#)

Q10. Which HTTP verb is normally used to update or create a resource in an API?

- ☐ SUBMIT
- ☐ WRITE
- ☒ POST
- ☐ CREATE

[HTTP request methods](#)

Q11. What is one benefit of server-side caching in APIs?

- ☐ Mobile app work better.
- ☐ It improves uptime.
- ☐ It offers better security.
- ☒ It reduce load on servers.

Q12. Your API resource does no allow deletion, and a client application attempted to delete the resource. What HTTP response code should you return?

- ☐ 409 Conflict
- ☐ 400 Bad Request
- ☐ 406 Not Acceptable
- ☒ 405 Method Not Allowed

[Response Codes](#)

Q13. What is OpenID Connect?

- ☒ an identify layer on top of OAuth 2.0
- ☐ the new name for SAML 3.0
- ☐ a modern replacement for API keys
- ☐ an SSO competitor for OAuth 2.0

[What is OpenID Connect?](#)

Q14. What is one benefit of GraphQL over REST approaches?

- ☒ flexible querying/responses
- ☐ more stable APIs
- ☐ compatible with more gateways
- ☐ more secure by default

[GraphQL vs. REST](#)

Q15. Which REST constraint specifies that there should be no shared context?

- ☒ Stateless
- ☐ Client-Server
- ☐ Uniform Interface

- ☐ Cacheable

Q16. What purpose does a User-Agent serve?

- ☐ It identifies the user ID.
- ☒ It identifies the client application or SDK.
- ☐ It identifies if the API should expect a user authentication.
- ☐ It identifies if the API should accept microservice traffic.

Q17. If you were to add versioning by using the Accept and Content-Type header, what would be the correct format of the header value?

- ☐ application/json
- ☐ application/json_version2
- ☐ text/html
- ☒ application/vnd.myapp.v2+json

[Reference](#)

Q18. What is one benefit that OAuth provides over an API key approach?

- ☐ A token is encrypted.
- ☐ A token is encoded.
- ☒ A token is scoped to the use case.
- ☐ A token can be shared between systems.

[How to easily secure your APIs with API keys and OAuth](#)

Q19. The ability to execute the same API request over and over again without changing the resource's state is an example of _.

- ☐ stateless architecture
- ☒ idempotency
- ☐ a uniform interface
- ☐ cacheability

Q20. What component can you use to wrap legacy architectures or protocols into a REST interface for easier consumption and integration?

- ☒ API proxy

- ☐ API gateway
- ☐ OpenAPI
- ☐ OAuth authorization server

Q21. What protection does a JSON Web Token (JWT) offer to mitigate tampering with its contents?

- ☐ transport over SSL
- ☐ encrypted payload
- ☒ a signature
- ☐ encoded payload

Q22. What OAuth term is used to represent permissions?

- ☐ token
- ☒ scope
- ☐ claim
- ☐ back channel

[OAuth Scopes](#)

Q23. What additional type of token would you see when using OpenID Connect?

- ☒ ID token
- ☐ refresh token
- ☐ access token
- ☐ auth code token

Q24. What should you add to a Cache-Control response header to specify that a response should not be stored in an intermediary cache?

- ☐ no-proxy
- ☐ client-only
- ☐ restricted
- ☒ private

[reference](#)

Q25. Which OAuth grant type can support a refresh token?

- ☒ Authorization Code Grant
- ☐ Client Credentials Grant
- ☐ Implicit Grant
- ☐ Authentication Grant

[Reference:](#)

Q26. Using OAuth, what scope would you request for write access to the API?

- ☐ It varies from API to API.
- ☒ admin
- ☐ write
- ☐ read-write

Q27. Which property would you use to include subresources directly into a JSON document?

- ☐ _embedded
- ☐ resources
- ☒ subresources
- ☐ _links

Q28. What is the best way to track SDK and version usage?

- ☒ tracking downloads
- ☐ Accept headers
- ☐ user agents
- ☐ polling users

Q29. Which REST constraint allows for the presence of caching, routing, and other systems between the client and server?

- ☒ Layered System
- ☐ Stateless
- ☐ Client-Server
- ☐ Cacheable

Q30. Which content is best to include in your documentation?

- ☐ your tech stack

- ☐ reasoning for your naming schema
- ☐ your mission statement
- ☒ sample code

Q31. What metric tracks overall availability for your API?

- ☐ Response Time
- ☐ Time to First Hello World
- ☐ TTL
- ☒ Uptime

Q32. What is the recommended method and URL pattern for retrieving a specific user?

- ☐ GET /user/{id}
- ☒ GET /users/{id}
- ☐ GET /user?id={id}
- ☐ GET /users?id={id}

Q33. What is the purpose of a link relation?

- ☐ to describe relationships between resources or actions
- ☐ to describe subresources related to the current one
- ☒ to link two resources together
- ☐ to describe a resource and its purpose

Q34. When building SDKs, which languages should you support?

- ☐ Java, Javascript, and .NET
- ☐ and you can support
- ☐ PHP, Python, and Go
- ☒ the languages that your target users use

Q35. Which property would you use to include references to other resources in a JSON document?

- ☒ resources
- ☐ _embedded
- ☐ subresources

- ☐ `_links`

Q36. What is OAuth?

- ☒ an authorization framework for granted delegated access
- ☐ an approach to single sign-on for APIs
- ☐ a method for API authentication
- ☐ HTTP Basic Authentication 2.0

Q37. What should your API documentation describe?

- ☐ JSON
- ☐ HTTP
- ☒ common use cases
- ☐ your tech stack

Q38. What is the purpose of an OAuth refresh token?

- ☐ to share user profile information
- ☐ to update an API configuration
- ☐ to keep a web session active
- ☒ to retrieve an access token

[Understanding Refresh Tokens](#)

Q39. What is Time to First Hello World?

- ☒ how long it takes for a developer to do something with your API
- ☐ how long it takes to start a new programming language
- ☐ how long it takes to install your SDK
- ☐ how long it takes to read your documentation

Q40. Which response header tells the client and intermediaries that the response is not to be cached anywhere?

- ☐ Cache-State: none
- ☐ Expires:-1

- ☐ Cache-Control: no-cache
- ☒ Cache-Control: no-store

Q41. What component hides the distinctions or boundaries between various microservices from end-client applications?

- ☒ API gateway
- ☐ API logging
- ☐ a layered system
- ☐ API proxy

Q42. The textbook approach to api versioning is to use _.

- ☐ common knowledge
- ☐ URLs
- ☐ no versioning
- ☒ the Accept header

Q43. Which is the most secure method to transmit an API key?

- ☐ URL parameter
- ☒ Authorization header
- ☐ Base64 encoding
- ☐ Basic Auth

Q44. Within Oauth, what component validates the user's identity?

- ☐ client
- ☐ not specified
- ☒ authorization server
- ☐ resource server

Q45. API traffic that is entirely internal to your organization is normally called _?

- ☐ inbound traffic
- ☐ north-south traffic
- ☒ internal traffic
- ☐ east-west traffic

Q46. What is the best approach for requesting JSON instead of XML from an API?

- ☐ Add .json to the URL.
- ☐ APIs do not use XML.
- ☐ Use the Content-Type header.
- ☒ Use the Accept header.

Q47. When a user attempts to access a record that is not their own, which HTTP response code is the most appropriate?

- ☒ 403
- ☐ 404
- ☐ 401
- ☐ 405

Response Codes

Q48. Which is a benefit of using an API gateway?

- ☐ HTTP verbs
- ☐ JSON payloads
- ☐ HTTP response codes
- ☒ rate limiting/throttling

Q49. API testing must be treated as _?

- ☐ red team testing
- ☐ white box testing
- ☐ blue box testing
- ☒ black box testing

Q50. Which HTTP verb is used in a CORS preflight request?

- ☐ PUT
- ☐ POST
- ☐ GET
- ☒ OPTIONS

Q51. Which response header will tell the client that the response is cached for 1 minute ?

- ☐ Expires: 1 minute
- ☒ Cache-Control: max-age=60
- ☐ Expires: 1 January 2020
- ☐ Cache-Expires: max-age=60

[Cache Control Header](#)

Q52. What is the concept that allows an API client to explore an API via links embedded in payloads?

- ☐ hypermedia
- ☒ link relations
- ☐ parsing
- ☐ browsing

Q53. To create a new resource, what HTTP response code should you receive?

- ☐ 405
- ☒ 201
- ☐ 204
- ☐ 202

[Response Codes](#)

Q54. Which is an example of Code on Demand?

- ☐ AWS Lambda
- ☐ downloading open-source software
- ☐ Serverless
- ☒ JavaScript on a webpage

[Code on Demand](#)

Q55. Which URL pattern should you follow for accessing a subresource attached to a specific resource?

- ☐ /companies/employees/{companyId}/{employeeId}

- ☐ /company/{companyId}/employees/{employeeId}
- ☒ /companies/{companyId}/employees/{employeeId}
- ☐ /companies/{companyId}/employee/{employeeId}

Resource Naming

Q56. Which REST constraint essentially prohibits the use of cookies?

- ☒ Stateless
- ☐ Cacheable
- ☐ Layered System
- ☐ Uniform Interface

Q57. Which HTTP verb is used to delete a resource?

- ☐ FLUSH
- ☒ DELETE
- ☐ CLEAR
- ☐ DESTROY

Q58. Which verb is *not* considered idempotent?

- ☐ DELETE
- ☐ GET
- ☐ PUT
- ☒ POST

Idempotency

Q59. Which REST constraint specifies that each request should stand on its own and not have a specific required order?

- ☐ Uniform Interface
- ☐ Cacheable
- ☒ Stateless
- ☐ Client-Server

REST Architectural Constraints

Q60. When you get a 429 response code, what should you do next?

- ☐ Check you JSON structure.
- ☒ Slow down your requests.
- ☐ Check the API uptime status.
- ☐ Check you API key.

Q61. When exploring record sets, what is the best approach for pagination?

- ☐ date-based filtering
- ☒ next/previous cursors
- ☐ page size and filters
- ☐ database IDs

Q62. What is *not* a method for API authentication or authorization?

- ☐ OAuth
- ☒ biometrics
- ☐ API Keys
- ☐ username and password

Q63. Which HTTP response code usually means the requested work is still processing and may or may not result in an error later?

- ☐ 200 OK
- ☐ 204 No Content
- ☐ 201 Created
- ☒ 202 Accepted

Q64. When validating a JWT, what are some of the claims that you must confirm? (Select all that apply.)

- A. The exp (expiration) has not passed.
- B. The algorithm is sufficient.
- C. The signature matches the payload.
- D. The token was Base64 encoded.
- E. The iss (issuer) is the auth server you expect.
- F. There is a refresh token.
- G. The cid (client ID) is the client you expect.
- H. The token was encrypted.

- ☒ A,B,E,H
- ☐ B,C,F,G
- ☐ A,D,G,H
- ☐ A,C,E,G

Q65. API traffic that enters and leaves your organization is normally called _?

- ☐ east-west traffic
- ☐ inbound traffic
- ☒ north-south traffic
- ☐ external traffic

[North-South vs East-West Traffic](#)

Q66. Which OAuth grant type is appropriate for mobile apps?

- ☒ Authorization Code with PKCE
- ☐ Client Credentials
- ☐ Device
- ☐ Resource Owner Password

[OAuth 2.0 for Native and Mobile Apps](#)

Q67. Which datetime format is the easiest or most predictable to parse and process?

- ☐ YY-M-D hh:mm:ss+TZ
- ☐ YY-M-D h:mm:ss
- ☒ YYYY-MM-DDThh:mm:ssZ
- ☐ YYYY-M-D hh:mm:ss

[The 5 laws of API dates and times](#)

Q68. Which header is *not* used in cache management?

- ☒ Rate-Limit
- ☐ Expires
- ☐ Etag
- ☐ Cache-Control

Cache-Control Expires Etag Rate limiting your RESTful API

Q69. A client application uses a filter or a search in your API correctly but there are zero results. What is the best response code?

- ☐ 204 No Content
- ☐ 400 Bad Request
- ☒ 200 OK
- ☐ 404 Not Found